

KAIST
2023 Fall
[MAS456] Statistical Methods with Computer
Homework 5

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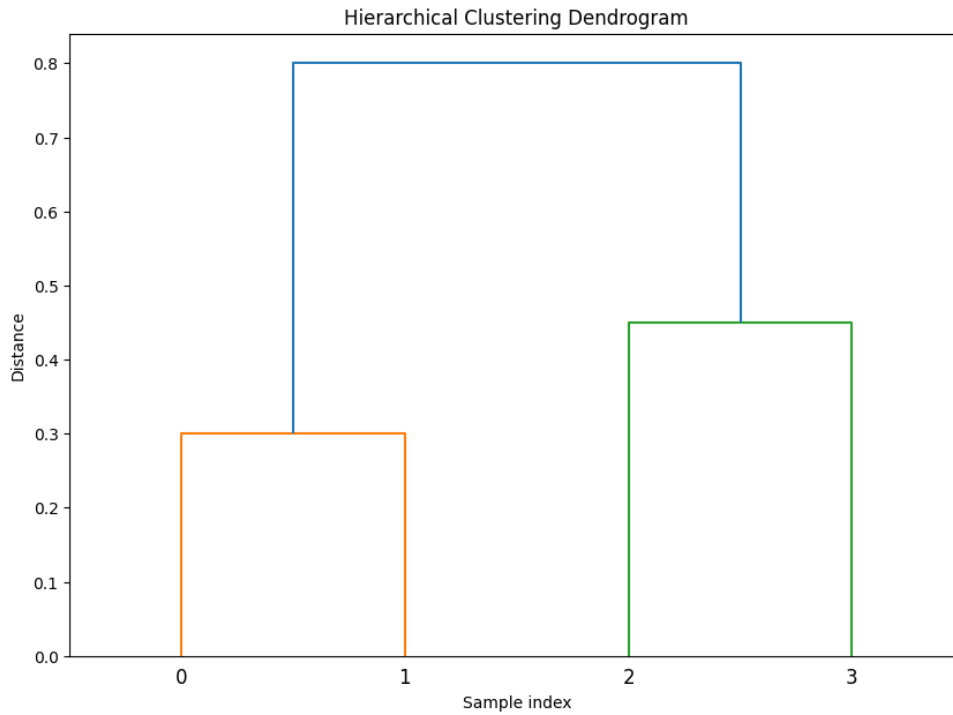
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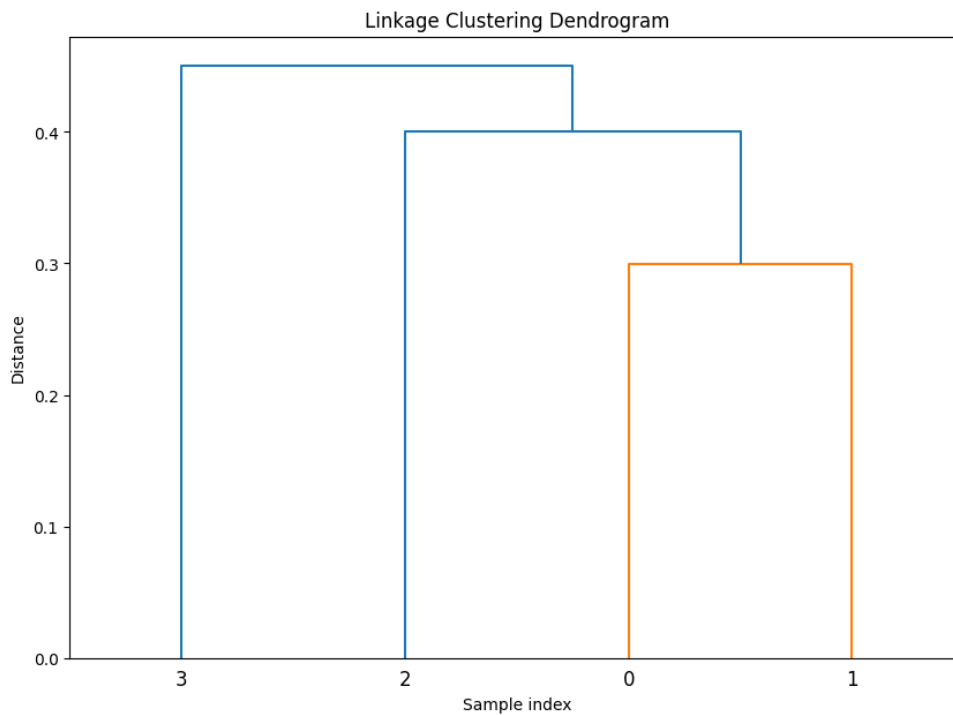
Chapter 12 - Unsupervised Learning

Problem #2

(a) Dendrogram:



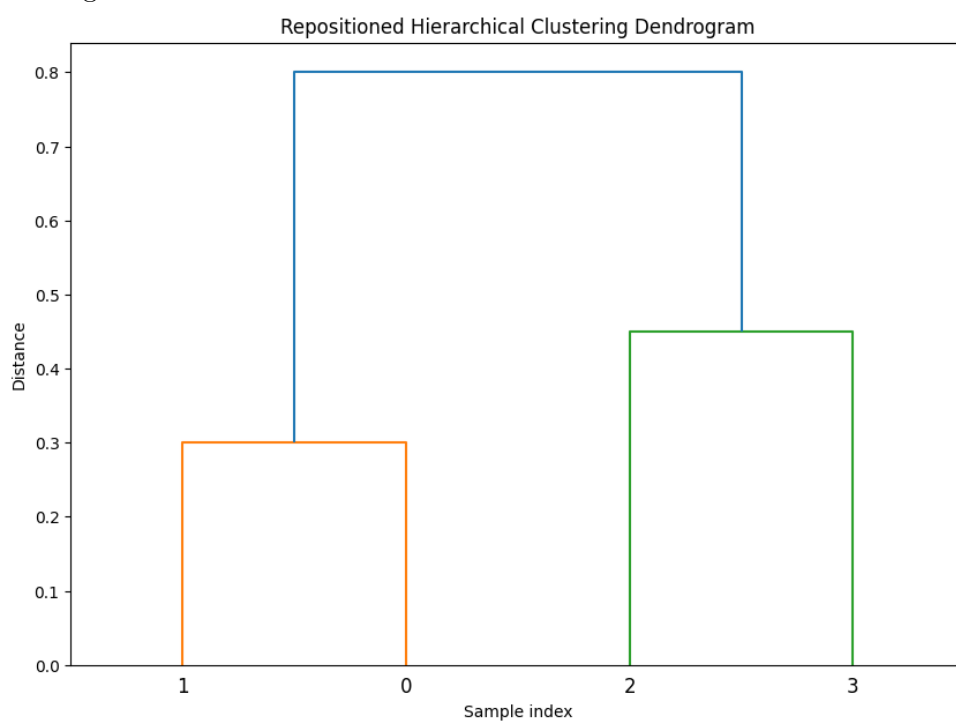
(b) Dendrogram:



(c) Observation 0 and 1 in is one cluster. Observation 2 and 3 is in another cluster. [0-indexed]
 Cluster 1: $[0, 1]$, Cluster 2: $[2, 3]$

(d) Cluster 1: $[0, 1, 2]$, Cluster 2: $[3]$

(e) Dendrogram:



Problem #3

(a) Plot the Observations

The plot of the observations is shown in Figure 1.

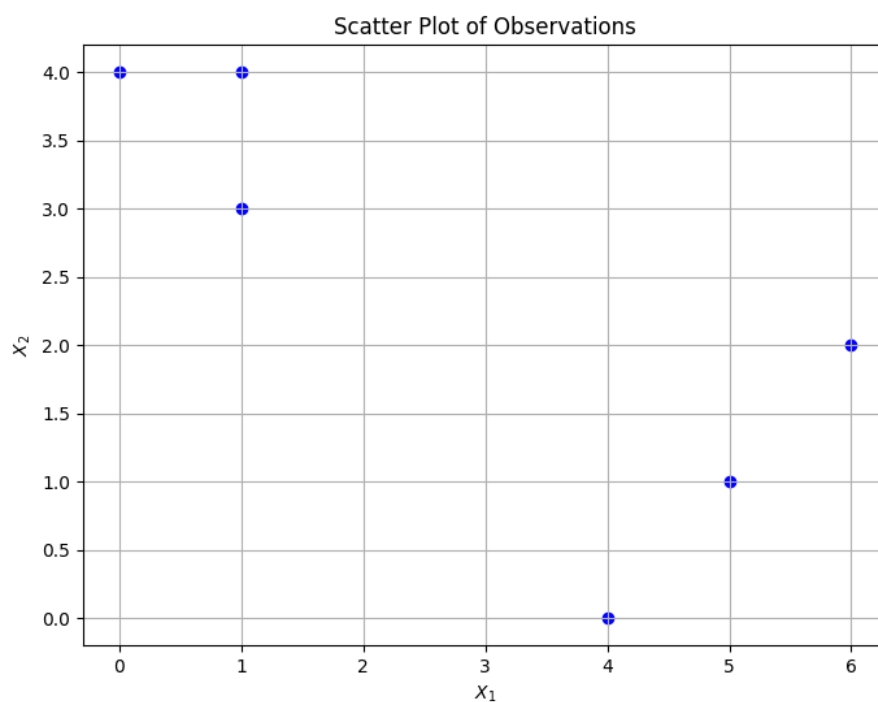


Figure 1: Plot of Observations

(b) Randomly Assign Cluster Labels

Using `np.random.choice()`, the observations are randomly assigned to two clusters. The assigned labels are as follows:

Obs.	Cluster Label
1	1
2	2
3	1
4	1
5	1
6	2

(c) Compute the Centroids

The centroids for each cluster are calculated as follows:

For Cluster 1:

$$\begin{aligned}\text{Centroid} &= \left(\frac{\text{Sum of } X_1 \text{ values in Cluster 1}}{\text{Number of observations in Cluster 1}}, \frac{\text{Sum of } X_2 \text{ values in Cluster 1}}{\text{Number of observations in Cluster 1}} \right) \\ &= \left(\frac{1 + 0 + 5 + 6}{4}, \frac{4 + 4 + 1 + 2}{4} \right) \\ &= (3.00, 2.75)\end{aligned}$$

For Cluster 2:

$$\begin{aligned}\text{Centroid} &= \left(\frac{1 + 4}{2}, \frac{3 + 0}{2} \right) \\ &= (2.50, 1.50)\end{aligned}$$

(d) Reassign Observations to Closest Centroid

Observations are reassigned to the closest centroid based on Euclidean distance. The updated cluster labels are:

Obs.	Cluster Label
1	1
2	1
3	1
4	2
5	1
6	2

(e) Continuously updating centroids and cluster labels

After several iterations, the clusters stabilize. The final cluster labels are shown in the table above, and the corresponding plot is in Figure 2.

Obs.	Cluster Label
1	1
2	1
3	1
4	2
5	2
6	2

(f) Final Clusters and Plot

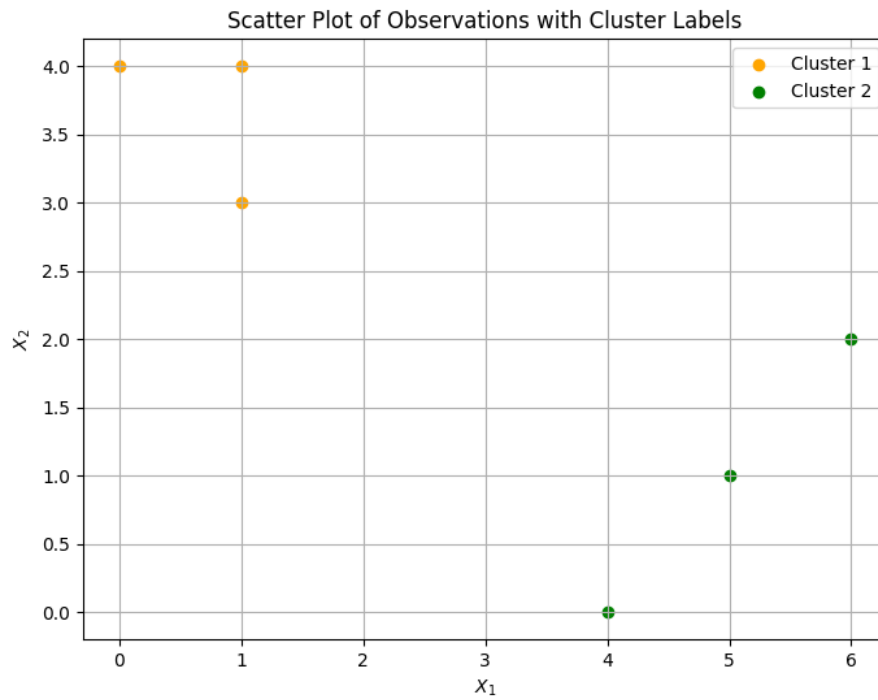


Figure 2: Final Clustering

Problem #8

(a) PVE Using `explained_variance_ratio_`

The PVE is calculated using the `explained_variance_ratio_` attribute of the fitted PCA estimator. The calculated PVE is:

$$\text{PVE} = [0.62006039, 0.24744129, 0.0891408, 0.04335752]$$

(b) PVE Using Equation 12.10

Using the loadings from the `components_` attribute of the fitted PCA estimator, the PVE is calculated as:

$$\text{PVE calculated} = [0.62006039, 0.24744129, 0.0891408, 0.04335752]$$

Problem #9

(a) Hierarchical Clustering with Complete Linkage

Hierarchical clustering with complete linkage and Euclidean distance was performed. The resulting dendrogram is shown in Figure 3.

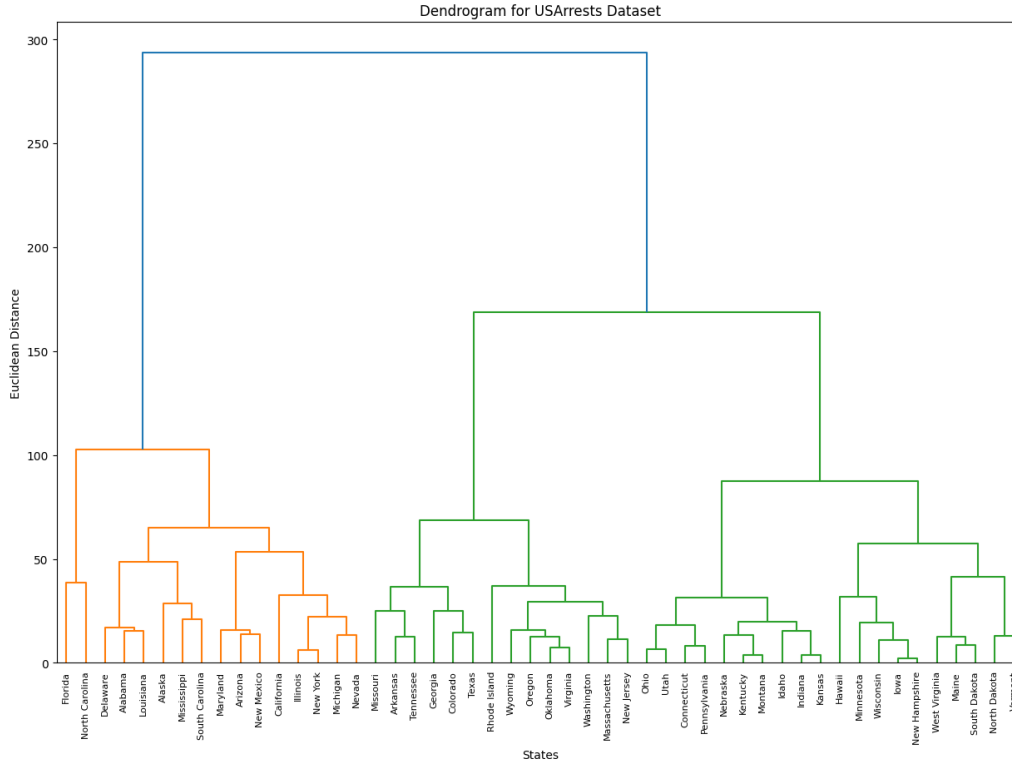


Figure 3: Dendrogram for Hierarchical Clustering

(b) Three Distinct Clusters

The dendrogram was cut at a height that resulted in three distinct clusters. The states belonging to each cluster are as follows:

- Cluster 1: ['Alabama', 'Alaska', 'Arizona', 'California', 'Delaware', 'Florida', 'Illinois', 'Louisiana', 'Maryland', 'Michigan', 'Mississippi', 'Nevada', 'New Mexico', 'New York', 'North Carolina', 'South Carolina']
- Cluster 2: ['Arkansas', 'Colorado', 'Georgia', 'Massachusetts', 'Missouri', 'New Jersey', 'Oklahoma', 'Oregon', 'Rhode Island', 'Tennessee', 'Texas', 'Virginia', 'Washington', 'Wyoming']
- Cluster 3: ['Connecticut', 'Hawaii', 'Idaho', 'Indiana', 'Iowa', 'Kansas', 'Kentucky', 'Maine', 'Minnesota', 'Montana', 'Nebraska', 'New Hampshire', 'North Dakota', 'Ohio', 'Pennsylvania', 'South Dakota', 'Utah', 'Vermont', 'West Virginia', 'Wisconsin']

(c) Hierarchical Clustering with Scaled Variables

The variables were scaled to have a standard deviation of one before performing hierarchical clustering. The dendrogram for this clustering is shown in Figure 4.

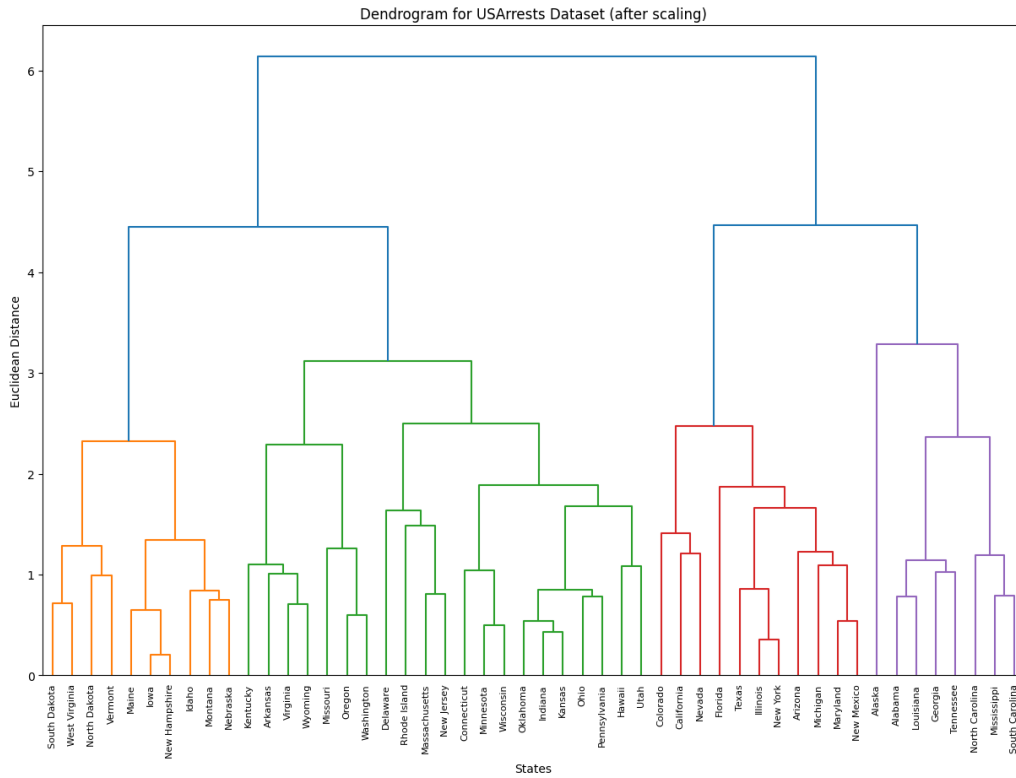


Figure 4: Dendrogram for Hierarchical Clustering with Scaled Variables

(d) Effect of Scaling on Hierarchical Clustering

Scaling the variables can significantly impact the results of hierarchical clustering. When variables are on different scales, those with larger scales can dominate the distance measure. Scaling ensures that each variable contributes equally to the distance calculations. In my opinion, scaling should be performed when variables are measured in different units or have very different ranges. This ensures that the clustering is not biased towards variables with larger scales.

In this case:

- Murder Murder arrests (per 100,000)
- Assault Assault arrests (per 100,000)
- UrbanPop Percent urban population
- Rape Rape arrests (per 100,000)

All of the variables are not on the same unit. So, it would be safer to scale the data so that each variable has equal weight.

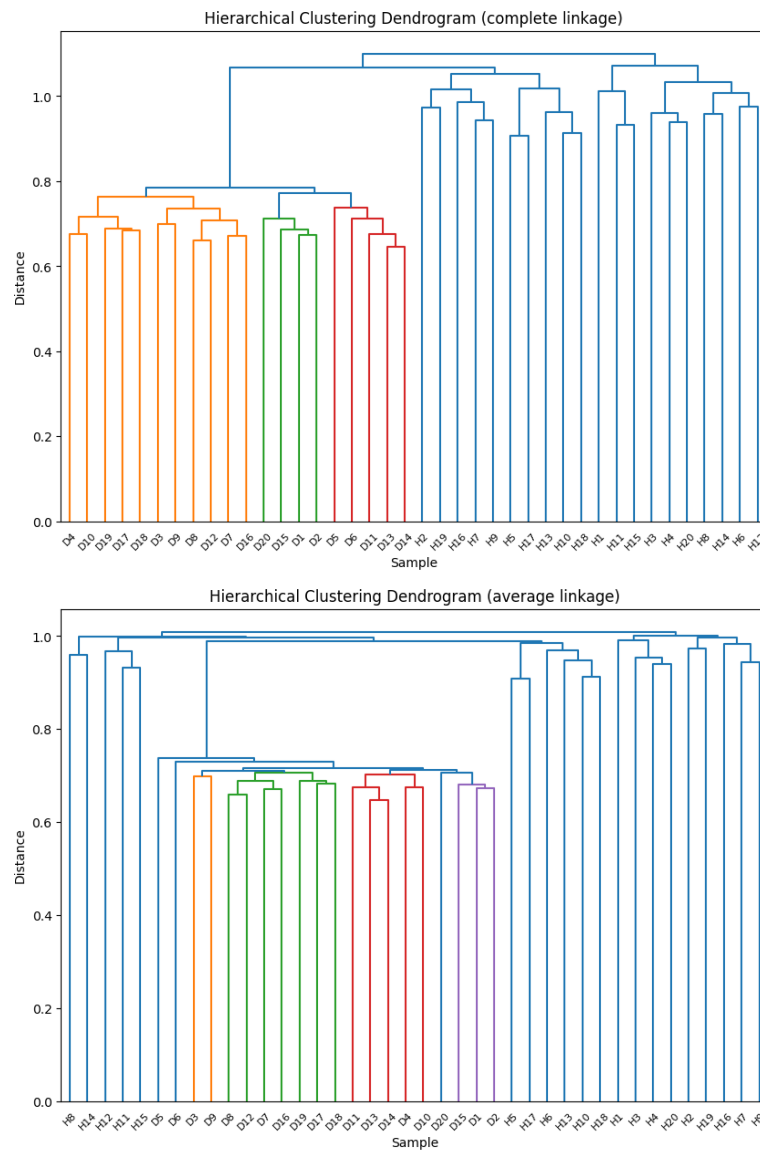
Problem #13

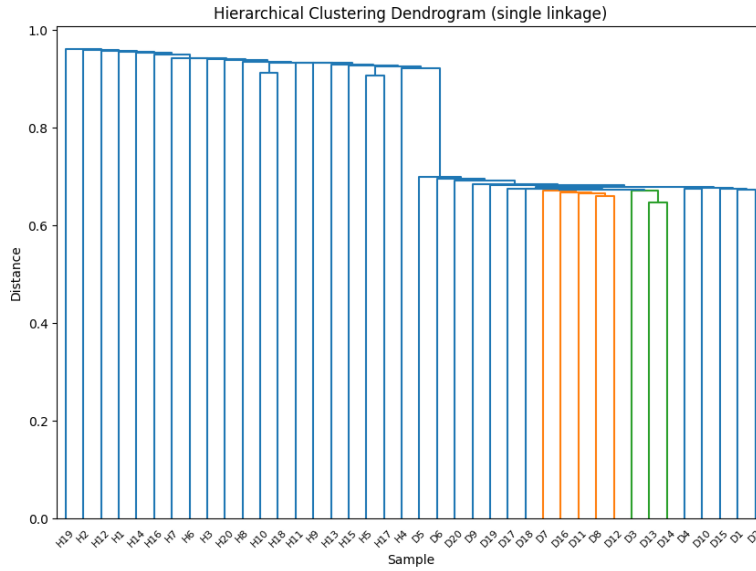
(a) Loading the Data

The data was loaded using the `pd.read_csv()` function in Python, specifying `header = None` to accommodate the dataset format.

(b) Hierarchical Clustering

Hierarchical clustering was performed using a correlation-based distance measure. The resulting dendrogram is shown in Figure 3.





The dendrogram was analyzed to determine if the genes effectively separate the samples into healthy and diseased groups. Different linkage methods (complete, average, single) were also explored to assess their impact on the clustering results.

What we have found out is that they clearly separate into two groups. However, how they are separated depends on the type of linkage used. While for the complete and single method, the separation was into two separate sets, for the average method, all the disease samples became a subset of the healthy samples.

(c) Identifying Differentially Expressed Genes

To identify genes that differ significantly between the healthy and diseased groups, the following approach was used:

- Compute the mean expression level of each gene in both groups.
- Use statistical tests (e.g., t-test) to identify genes with significantly different expression levels between the groups.
- Adjust for multiple testing (e.g., using the Bonferroni correction or false discovery rate methods) to control for Type I errors.

This method was applied to the dataset, and a list of genes showing the most significant differences between the two groups was compiled.

[11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 501, 502, 503, 504, 505, 506, 507, 508, 509, 511, 512, 513, 514, 515, 516, 517, 519, 520, 521, 522, 523, 524, 525, 526, 527, 528, 529, 530, 531, 532, 533, 534, 535, 536, 537, 538, 539, 540, 541, 542, 543, 544, 545, 546, 547, 548, 549, 550, 551, 552, 553, 554, 555, 556, 557, 558, 559, 560, 561, 562, 563, 564, 565, 566, 567, 568, 569, 570, 571, 572, 574, 575, 576, 577, 578, 579, 580, 581, 582, 583, 584, 586, 587, 588, 589, 590, 591, 592, 593, 595, 596, 597, 598, 599, 600]

Chapter 13 - Multiple Testing

Problem #1

(a) Number of Type I Errors Expected

Given m null hypotheses and a Type I error rate α for each test, the expected number of Type I errors is:

$$\text{Expected Type I errors} = m \times \alpha$$

(b) Family-wise Error Rate for Independent Tests

For m independent tests, the family-wise error rate (FWER) is calculated as:

$$\text{FWER} = 1 - (1 - \alpha)^m$$

(c) FWER with Positively Correlated P-values ($m = 2$)

When the p-values are positively correlated, the FWER is higher than in the independent case. If p-values are perfectly correlated, the FWER equals α .

(d) FWER with Negatively Correlated P-values ($m = 2$)

With negatively correlated p-values, the FWER is lower than in the independent case. In the extreme scenario where one p-value being less than α ensures the other is more than α , the FWER effectively becomes the probability of a single test, which is α .

Problem #2

(a) Distribution of A_j

Let A_j be a random variable such that $A_j = 1$ if the j -th null hypothesis is rejected, and 0 otherwise. Since the Type I error rate is α and all null hypotheses are true, A_j follows a Bernoulli distribution with parameter α . Thus:

$$A_j \sim \text{Bernoulli}(\alpha)$$

(b) Distribution of $\sum_{j=1}^m A_j$

The sum $\sum_{j=1}^m A_j$ represents the total number of rejected null hypotheses (Type I errors). Since each A_j is a Bernoulli distributed random variable and the tests are independent, the sum follows a Binomial distribution with parameters m and α . Therefore:

$$\sum_{j=1}^m A_j \sim \text{Binomial}(m, \alpha)$$

(c) Standard Deviation of the Number of Type I Errors

The standard deviation of a Binomial(m, α) distribution is given by:

$$\sigma = \sqrt{m\alpha(1 - \alpha)}$$

Thus, the standard deviation of the number of Type I errors is:

$$\sigma_{\text{Type I}} = \sqrt{m\alpha(1 - \alpha)}$$

Problem #4

(a) Type I Error Control at $\alpha = 0.05$

We reject those null hypotheses for which the p-value is less than $\alpha = 0.05$. The rejected null hypotheses are: H01, H02, H03, H08, H09, and H10.

(b) FWER Control at $\alpha = 0.05$

To control the FWER at level $\alpha = 0.05$, we can use the Bonferroni correction, dividing α by m . The corrected α is $0.05/10 = 0.005$. The null hypotheses rejected are those with p-values less than 0.005: H01, H09, and H10.

(c) FDR Control at $q = 0.05$

To control the FDR at level $q = 0.05$, we can use the Benjamini-Hochberg procedure. After ranking the p-values, we find the largest k such that $p(k) \leq \frac{k}{m}q$. The null hypotheses rejected are: H01, H03, H08, H09, and H10.

(d) FDR Control at $q = 0.2$

Repeating the Benjamini-Hochberg procedure for $q = 0.2$, we find that the null hypotheses rejected are: H01, H02, H03, H05, H07, H08, H09, and H10.

(e) False Positives at FDR Level $q = 0.2$

Under the FDR level of $q = 0.2$, we expect that approximately 20% of the rejected hypotheses are false positives. Since we rejected 8 hypotheses, the expected number of false positives is $0.2 \times 8 = 1.6$. Therefore, we expect approximately 1 or 2 false positives.

Problem #6

(a) Bonferroni Procedure at $\alpha = 0.05$

The Bonferroni procedure's application at $\alpha = 0.05$ yields the following results for false positives (FP), false negatives (FN), true positives (TP), true negatives (TN), Type I and Type II errors across the three panels:

Panel	FP	FN	TP	TN	Type I	Type II
1	0	1	7	2	0	1
2	0	1	7	2	0	1
3	0	5	3	2	0	5

Table 1: Results using the Bonferroni procedure.

(b) Holm Procedure at $\alpha = 0.05$

Using the Holm procedure to control FWER at the same level results in:

Panel	FP	FN	TP	TN	Type I	Type II
1	0	1	7	2	0	1
2	0	0	8	2	0	0
3	0	0	8	2	0	0

Table 2: Results using the Holm procedure.

(c) False Discovery Rate with Bonferroni Procedure

The false discovery rate (FDR) is the expected ratio of false positives to total positives. Under the Bonferroni procedure at $\alpha = 0.05$, the FDR for all panels is 0, as there are no false positives.

(d) False Discovery Rate with Holm Procedure

Similarly, with the Holm procedure at $\alpha = 0.05$, the FDR remains 0 for all panels due to the absence of false positives.

(e) Bonferroni Procedure at $\alpha = 0.001$

Applying the Bonferroni procedure at a more stringent threshold of $\alpha = 0.001$ does not lead to any additional false positives. Therefore, the outcomes for both (a) and (c) remain unchanged.

Problem #8

(a) One-Sample t-Tests and Histogram of p-Values

Each fund manager's returns were subjected to a one-sample t-test. The p-values obtained from these tests were used to assess the significance of the results.

(b) Type I Error Control

When controlling for Type I error at a level of $\alpha = 0.05$, 3 null hypotheses were rejected.

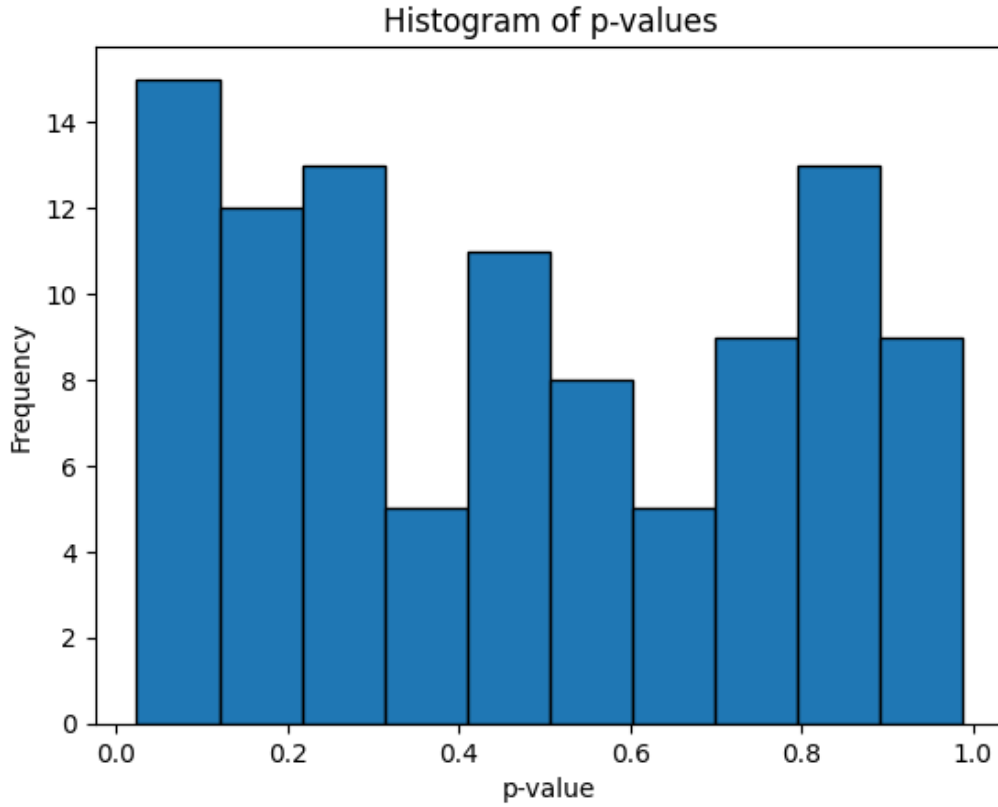


Figure 5: Histogram of p-values from one-sample t-tests.

(c) Family-Wise Error Rate (FWER) Control

Under the control of the Family-Wise Error Rate (FWER) at a level of 0.05, 0 null hypotheses were rejected.

(d) False Discovery Rate (FDR) Control

When controlling the False Discovery Rate (FDR) at a level of 0.05, 0 null hypotheses were rejected.

(e) Cherry-Picked Analysis

For the top 10 performing fund managers:

- Under FWER control at level 0.05: 0 null hypotheses were rejected.
- Under FDR control at level 0.05: 0 null hypotheses were rejected.

(f) Discussion on Cherry-Picked Analysis

The cherry-picked analysis of the top-performing fund managers is misleading due to the introduction of selection bias and the violation of assumptions inherent in standard error control procedures. Selecting fund managers based on their performance after observing the data increases the risk of Type I errors and does not reflect an unbiased assessment of their overall performance. This approach fails to consider the multiplicity of tests performed and selectively focuses on extreme outcomes, which may be more attributable to chance than to genuine performance.